
Insurance Charges Prediction

A machine learning regression study: from exploratory analysis to a tuned, validated final model.

PROJECT REPORT

Python · scikit-learn · XGBoost · pandas · matplotlib / seaborn

Final model: Random Forest with log-transformed target · Test RMSE 4,611 · Test MAE 2,132

1. Project overview

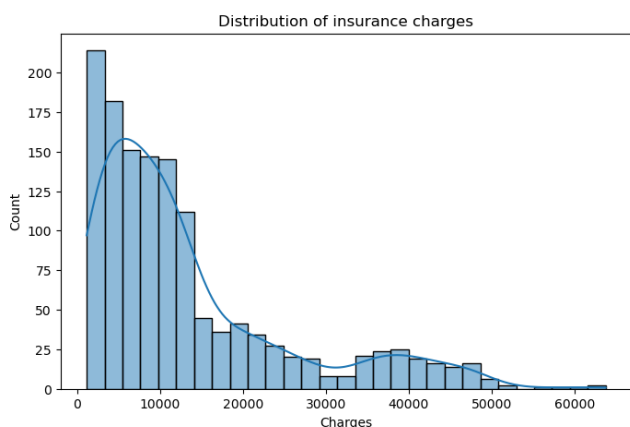
This project predicts individual annual medical insurance charges from six attributes: age, sex, BMI, number of children, smoker status and region. The dataset holds 1,338 records (1,337 after removing one duplicate) with no missing values. The target, **charges**, is continuous and strongly right-skewed, which shapes both the modelling choices and the evaluation.

The work follows a disciplined pipeline: clean the data, explore it, build a leakage-free preprocessing pipeline, compare five algorithms with cross-validation and hyperparameter search, address target skew with a log transform, and finally evaluate a single chosen model on a held-out test set that is used only once.

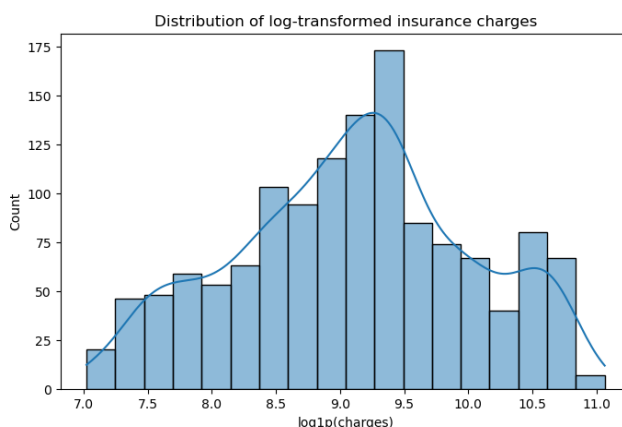
1,337 Records	6 Features	4,611 Test RMSE	2,132 Test MAE
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2. Exploratory data analysis

The target is right-skewed (skewness of about 1.52). A log transform, $\log_{1p}(\text{charges})$, pulls the long right tail in and produces a far more symmetric distribution, which is easier for models to fit.

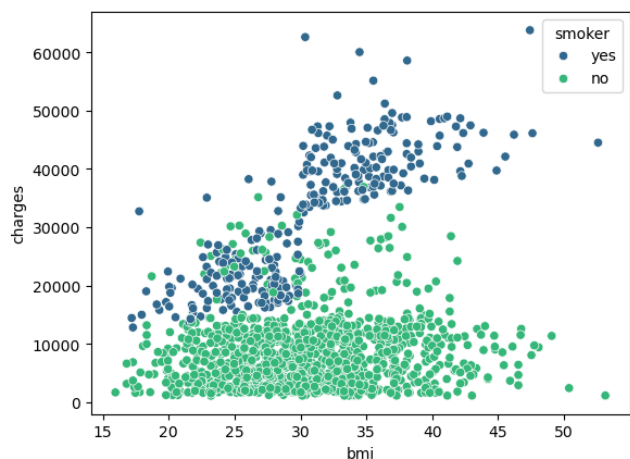


Raw charges: a long right tail of high-cost patients.

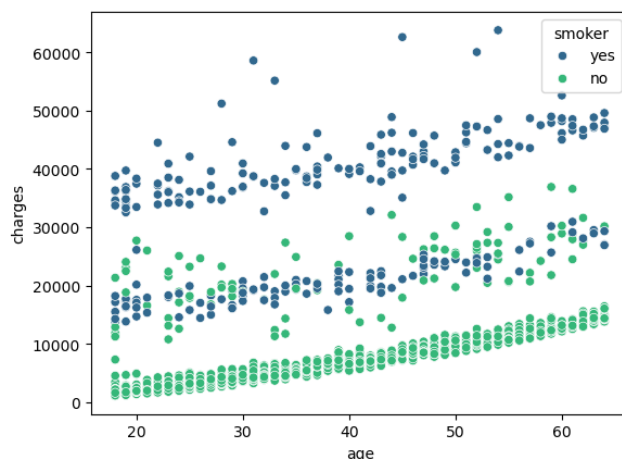


After log1p: a much more symmetric distribution.

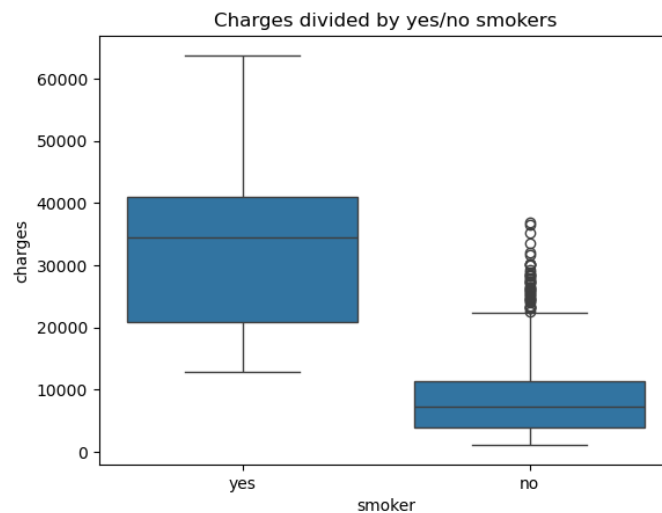
Splitting the data by smoker status is the single most revealing view. Smokers form a separate, far more expensive cluster at every BMI and every age. Charges also rise steadily with age in both groups. These two signals, smoking and age, drive most of the variation in cost.



BMI vs charges, split by smoker: two distinct cost clusters.



Age vs charges, split by smoker: cost rises with age.



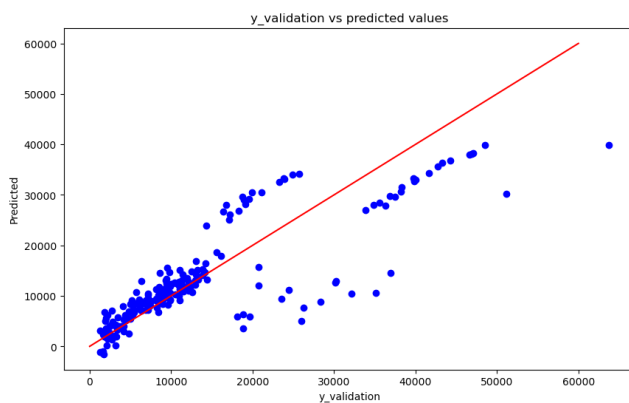
Charges by smoker status. The median smoker pays several times more than a non-smoker.

The highest-charge records all belong to obese smokers, which is medically plausible rather than a data error, so no outliers were removed.

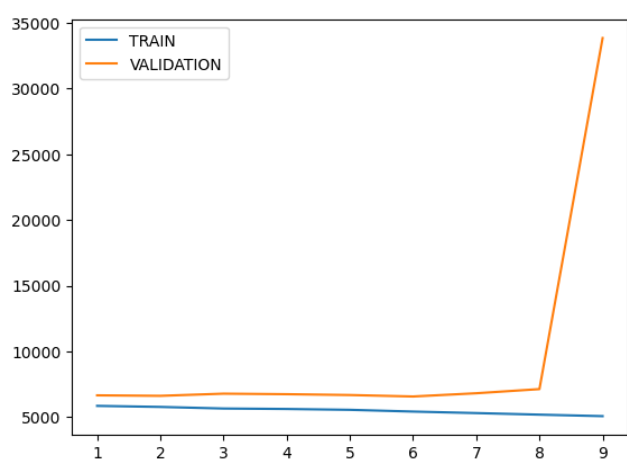
3. Preprocessing and modelling

Categorical features are one-hot encoded (dropping the first level) and numeric features are standardized, all inside a ColumnTransformer that is fit only on the training split. Validation and test data are transformed with the training statistics, so no information leaks into the model.

A plain Linear Regression baseline reaches a validation RMSE of about 6,651 against a mean charge of about 13,468. Its predicted-versus-actual plot fans out badly and the residuals show clear structure, confirming that a straight-line model underfits this problem. A polynomial-degree elbow study offered no meaningful gain over degree one, so added complexity alone was not the answer.



Linear baseline: predictions drift far from the ideal diagonal.



Elbow study: higher polynomial degrees do not help validation error.

4. Model comparison

Five algorithms were tuned with cross-validation and, because the target is skewed, wrapped in a TransformedTargetRegressor that trains on $\log_{1p}(\text{charges})$ and inverts predictions with expm1 . Random Forest on the log target gave the lowest validation RMSE and the smallest train-validation gap.

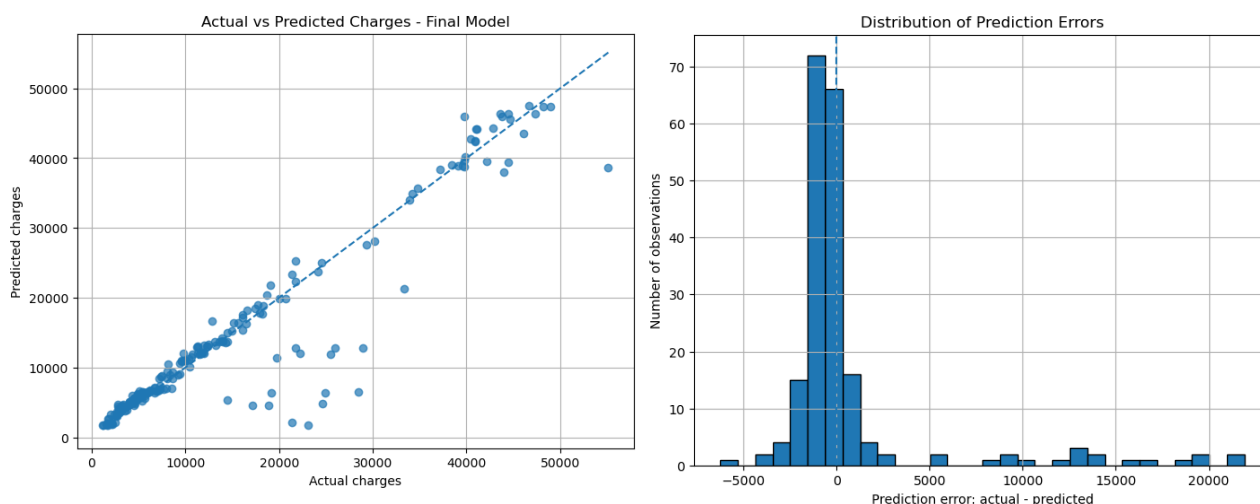
Model (log target)	Validation RMSE
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Random Forest	5,275
XGBoost	5,387
SVR	5,821
Linear Regression	7,702
ElasticNet	7,707

The plain Linear Regression baseline (no log target) scored a validation RMSE of about 6,651 for comparison.

5. Final model and results

The chosen model is a Random Forest Regressor wrapped in a TransformedTargetRegressor (log1p / expm1). Evaluated once on the held-out test set it reaches an RMSE of 4,611 and an MAE of 2,132. The MAE means the typical prediction is off by about \$2,132; the higher RMSE reflects a handful of high-cost cases that produce large errors, which is expected given the skewed target.



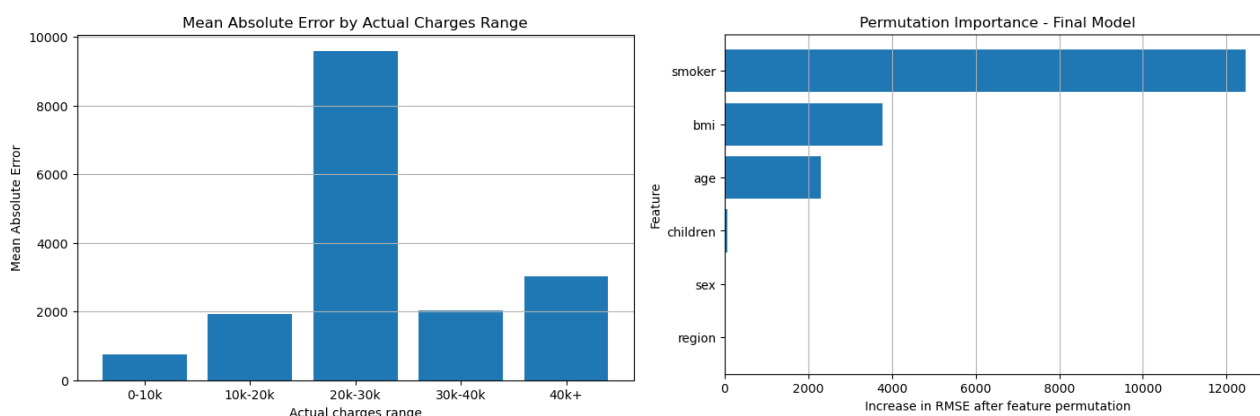
Test set: predictions track the ideal diagonal closely.

Prediction errors are centered near zero with a modest right tail.

6. Error analysis and feature importance

Breaking errors down by segment shows where the model struggles. Accuracy is strong at the low and high ends of the cost range, but the \$20k to \$30k band carries the largest errors: the model tends to under-predict these mid-cost patients, who sit between the low non-smoker cluster and the high smoker cluster.

Permutation importance confirms the story from the EDA, with smoker status dominating, followed by BMI and age, while children, sex and region add almost nothing.



Mean absolute error by actual charge band: the 20k to 30k range Permutation importance: smoker, then BMI and age, dominate. is hardest.

Takeaways. A strict train / validation / test protocol and leakage-free preprocessing mattered as much as the choice of algorithm. Matching the model to the data, by log-transforming a skewed target, produced a larger improvement than swapping algorithms. Segment-level error analysis revealed exactly where the model is weakest, which a single aggregate metric would have hidden.